

Samia Rahman

437-983-6021 | samia5@ualberta.ca | linkedin.com/in/samia | github.com/samia | samia-rahman.vercel.app

Education

University of Alberta

BSc Specialization in Computing Science

Edmonton, Alberta

Sept. 2022 – May 2027

Relevant Coursework: Software Development & Testing, Algorithms & Data Structures, Database Management, Artificial Intelligence, Machine Learning, Software Product Management, Data Science

Experience

Data Engineer Intern

May 2025 – December 2025

City of Edmonton – Financial & Corporate Services

Edmonton, AB

- Automated processing of millions of records for 500+ users by building scalable backend data pipelines integrating 20 SAP Ariba REST APIs using Python (Pandas, NumPy, SQLAlchemy) and SQL Server.
- Improved data reliability by implementing automated validation, monitoring, and data quality controls, reducing reporting discrepancies by 30% across supply chain systems.

Machine Learning Engineer Intern

Jan 2025 – Apr 2025

Process Data Analytics and Smart Automation Lab (PDASA)

Edmonton, AB

- Reduced manual bitumen extraction monitoring by 80% by deploying a PyTorch-based A3C reinforcement learning agent with CNN-LSTM encoders to automate closed-loop valve regulation via camera sensors.
- Reduced training time by 35% by migrating a legacy RL system from TensorFlow 1.x to PyTorch, redesigning training pipelines and model components.

ML & Robotics Research Assistant

Sep 2024 – Present

University of Alberta

Edmonton, AB

- Reduced training data requirements by 40% by developing PyTorch ML pipelines combining variational autoencoders (VAE) and Soft Actor-Critic (SAC) to improve policy generalization in robotic control environments.
- Improved UAV control performance by implementing and optimizing Model Predictive Control (MPC) algorithms, achieving 27% faster response, 41% increased stability, and 18% lower computational overhead.

Projects

UNTAPPED: Artist Management Web App | React, Node.js, PostgreSQL, OAuth2, AWS S3 Jan 2026 – Present

- Built a production artist management portal for ACPA Foundation enabling 50+ artists to submit invoices, track payments, and manage documents via a React frontend and Node.js/PostgreSQL backend.
- Eliminated manual admin workflows by implementing invoice approval pipelines, role-based access control, and PIPEDA-compliant AWS S3 document storage into a scalable multi-tenant architecture.

IoT Smart Home Test Automation | Java, JUnit 5, Docker, GitHub Actions

Dec 2025

- Achieved 80%+ branch coverage for state consistency logic (heating/cooling exclusivity, sensor accuracy, alarm triggers) by designing a TDD test suite using black-box and boundary value analysis in JUnit 5.
- Eliminated manual test execution by building a Docker + GitHub Actions CI/CD pipeline that automatically runs coverage analysis on every commit.

FoodMinder: Smart Grocery Management | TypeScript, React Native, Gemini API, Firebase

Feb 2025

- Built an LLM-powered React Native app using Google Gemini for receipt OCR and structured data extraction, reducing manual entry by 90% with real-time sync via Firebase.

Technical Skills

Full-Stack Development: Python, Java, C, TypeScript, JavaScript, SQL, R, React, React Native, Node.js, Express, HTML/CSS, REST APIs, PostgreSQL, MongoDB

AI/ML: Machine Learning, Reinforcement Learning, Large Language Models (LLMs), Generative AI, Prompt Engineering, RAG, Embeddings, PyTorch, TensorFlow, Scikit-learn

Data Engineering & Analytics: ETL Pipelines, Pandas, NumPy, Power BI, Tableau, Jupyter Notebook

Cloud & DevOps: AWS, Docker, Firebase, Linux, Git, GitHub Actions, CI/CD

Leadership

Ada's Team

May 2025 – present

VP Finance – Conferences

University of Alberta

- Managed \$45K+ budget and end-to-end conference logistics for Ada's Team, a diversity-in-tech club, securing industry sponsorships to fund travel and programming for 200+ underrepresented CS students across Edmonton